

STAT 412 QZ07 — Question 5

Problem

A sample of 49 batches of peanuts has mean aflatoxin level 24.74 ppm and variance 4.25 ppm².
Test

$$H_0 : \sigma^2 = 4.2$$

against

$$H_a : \sigma^2 \neq 4.2.$$

Use a P-value in the conclusion.

Solution

For a variance test,

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}.$$

Substitute $n = 49$, $s^2 = 4.25$, and $\sigma_0^2 = 4.2$:

$$\chi^2 = \frac{48(4.25)}{4.2} = 48.57.$$

With $df = 48$, the two-tailed P-value is

$$P \approx 0.900.$$

Since $0.900 > 0.05$, the result is not significant at the 0.05 level.

Final Answer

$\chi^2 = 48.57$; P-value = 0.900. The test statistic is not significant at the 0.05 level. Do not reject H_0 ; there is not sufficient evidence that $\sigma^2 \neq 4.2$.